Lecture 27 Sections 4.3 and 4.4

Section 4.3
Right Triangle Trigonometry

SOH CAH TOA

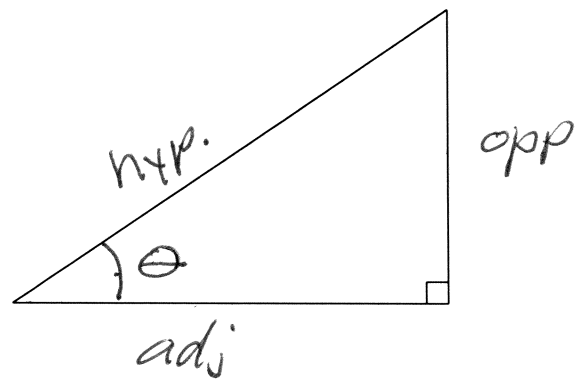
Cofunctions of complementary angles

Trigonometric identities

Section 4.4 Trigonometric Functions of Any Angle

Reference angle

Angle of elevation and angle of depression



Def. Let θ be an acute angle of a right triangle. The six trigonometric functions of the angle θ are defined as follows.

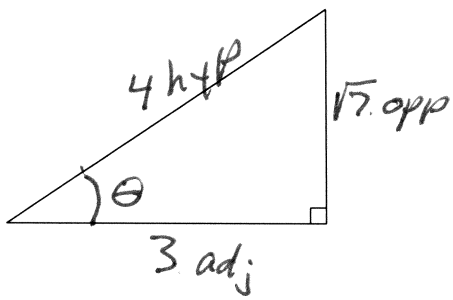
$$\sin\theta = \frac{opp}{hyp} \qquad \csc\theta = \frac{hyp}{opp}$$

$$\cos\theta = \frac{adj}{hyp} \qquad \sec\theta = \frac{hyp}{adj}$$

$$\tan\theta = \frac{opp}{adj} \qquad \cot\theta = \frac{adj}{opp}$$

SOH CAH TOA

ex. If $\cos\theta = \dfrac{3}{4}$, sketch a right triangle with acute angle θ, and find the other values of the five trigonometric functions of θ. $\cos\theta = \frac{3}{4} = \frac{adj}{hyp}$



$3^2 + ?^2 = 4^2$

$?^2 = 7$

$? = \sqrt{7}$

$\sin\theta = \frac{opp}{hyp} = \boxed{\frac{\sqrt{7}}{4}}$

$\csc\theta = \frac{hyp}{opp} = \frac{4}{\sqrt{7}}\frac{\sqrt{7}}{\sqrt{7}} = \boxed{\frac{4\sqrt{7}}{7}}$

$\cos\theta = \frac{adj}{hyp} = \boxed{\frac{3}{4}}$

$\sec\theta = \frac{hyp}{adj} = \boxed{\frac{4}{3}}$

$\tan\theta = \frac{opp}{adj} = \boxed{\frac{\sqrt{7}}{3}}$

$\cot\theta = \frac{adj}{opp} = \frac{3}{\sqrt{7}}\frac{\sqrt{7}}{\sqrt{7}} = \boxed{\frac{3\sqrt{7}}{7}}$

Checkpoint: Lecture 27, problem 1

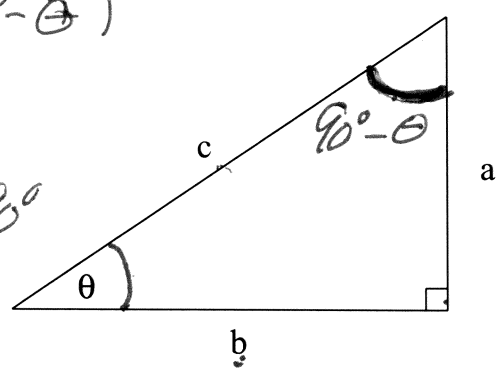
Theorem: Cofunctions of complementary angles are equal. That is,

$$\sin(90^\circ - \theta) = \cos\theta \qquad \cos(90^\circ - \theta) = \sin\theta$$

$$\tan(90^\circ - \theta) = \cot\theta \qquad \cot(90^\circ - \theta) = \tan\theta$$

$$\sec(90^\circ - \theta) = \csc\theta \qquad \csc(90^\circ - \theta) = \sec\theta$$

$\cos\theta = \frac{b}{c} = \sin(90^\circ - \theta)$



$\cos 70^\circ = \sin(90^\circ - 70^\circ)$

$\cos 70^\circ = \sin 20^\circ$

$70^\circ + 20^\circ = 90^\circ$

$90^\circ - \theta + \theta = 90^\circ$

$\cancel{\sec(\theta - 90^\circ) = \csc\theta}$

$\cancel{\theta - 90^\circ + \theta = 90^\circ}$

ex. Evaulate:

$$\begin{array}{r} 54 \\ 36 \\ \hline 90 \end{array}$$

1) $\sin 36^\circ - \cos 54^\circ$

$\cos 54^\circ - \cos 54^\circ = \boxed{0}$

$$\begin{array}{r} 56 \\ 34 \\ \hline 90^\circ \end{array}$$

2) $\dfrac{\tan 56^\circ}{\cot 34^\circ} = \dfrac{\cot 34^\circ}{\cot 34^\circ} = \boxed{1}$

Checkpoint: Lecture 27, problem 2

Trigonometric Identities

1. Reciprocal Identities

$$\sin\theta = \frac{1}{\csc\theta} \qquad \csc\theta = \frac{1}{\sin\theta}$$

$$\cos\theta = \frac{1}{\sec\theta} \qquad \sec\theta = \frac{1}{\cos\theta}$$

$$\tan\theta = \frac{1}{\cot\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$

2. Quotient Identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

3. Pythagorean Identities

$$\sin^2\theta + \cos^2\theta = 1$$

$\frac{1}{\sin^2\theta}\left[\sin^2\theta + \cos^2\theta = 1\right]$

$1 + \cot^2\theta = \csc^2\theta$

$\frac{1}{\cos^2\theta}\left[\sin^2\theta + \cos^2\theta = 1\right]$

$$1 + \tan^2\theta = \sec^2\theta$$

$\tan^2\theta + 1 = \sec^2\theta$

$$1 + \cot^2\theta = \csc^2\theta$$

ex. Let θ be an acute angle such that $\sin\theta = \frac{1}{3}$. Use trigonometric identities to find

1) $\cos\theta$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\left(\frac{1}{3}\right)^2 + \cos^2\theta = 1$$

$$1 - \frac{1}{9}$$

$$\cos^2\theta = \frac{8}{9}$$

$$\cos\theta = \sqrt{\frac{8}{9}} = \frac{\sqrt{8}}{3} = \boxed{\frac{2\sqrt{2}}{3}}$$

2) $\tan\theta$ Could use $\tan^2\theta + 1 = \sec^2\theta$

but quicker to use $\tan\theta = \frac{\sin\theta}{\cos\theta}$

$$\tan\theta = \frac{\frac{1}{3}}{\frac{2\sqrt{2}}{3}} = \frac{1}{\cancel{3}} \cdot \frac{\cancel{3}}{2\sqrt{2}} \frac{\sqrt{2}}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{4}}$$

Checkpoint: Lecture 27, problem 3

Lecture 27, Part II: Section 4.4
Trigonometric Functions of Any Angle

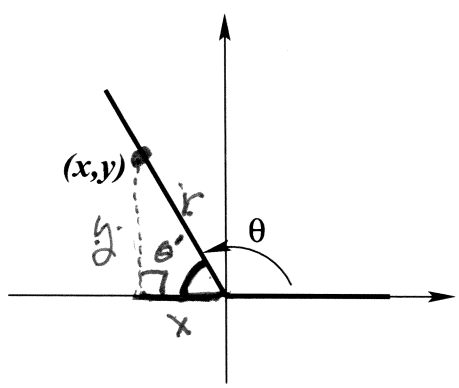
Not necessarily a unit circle, where $r=1$.

Def. Let θ be an angle in standard position and let (x, y) be a point on the terminal side. If $r = \sqrt{x^2 + y^2}$ is the distance from the origin to the point (x, y), then ~~work for any~~

$$\sin\theta = \frac{y}{r} \qquad \csc\theta = \frac{r}{y}$$

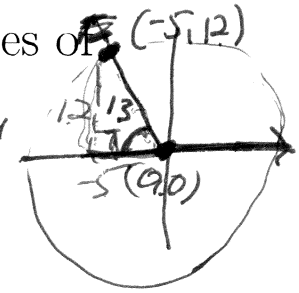
$$\cos\theta = \frac{x}{r} \qquad \sec\theta = \frac{r}{x}$$

$$\tan\theta = \frac{y}{x} \qquad \cot\theta = \frac{x}{y}$$



NOTE: If $x = 0$, $\tan\theta$ and $\sec\theta$ are undefined. If $y = 0$, $\cot\theta$ and $\csc\theta$ are undefined.

ex. Given point $(-5, 12)$ on the terminal side of the angle θ, find the six trigonometric function values of the angle θ. $r = d = \sqrt{(-5-0)^2 + (12-0)^2} = \sqrt{25+144}$

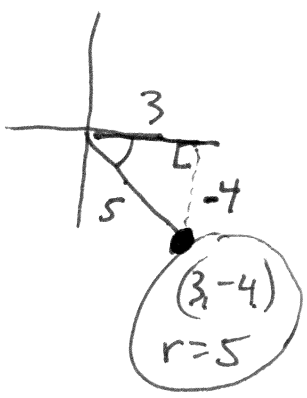


$\sin\theta = \frac{y}{r} = \boxed{\frac{12}{13}} = \frac{opp}{hyp}$ $r = 13$ $\csc\theta = \frac{r}{y} = \boxed{\frac{13}{12}}$

$\cos\theta = \frac{x}{r} = \boxed{\frac{-5}{13}} = \frac{adj}{hyp}$ $\sec\theta = \frac{r}{x} = \boxed{-\frac{13}{5}}$

$\tan\theta = \frac{y}{x} = \boxed{-\frac{12}{5}}$ $\cot\theta = \frac{x}{y} = \boxed{-\frac{5}{12}}$

ex. If $\cos\theta = \frac{3}{5}$ and θ lies in Quadrant IV, find the values of all the trigonometric functions.



$\cos\theta = \frac{x}{r} = \frac{3}{5}$

$\sin\theta = \frac{y}{r} = \boxed{\frac{-4}{5}} = -\frac{\cancel{5}}{\cancel{4}}$ $\csc\theta = \frac{r}{y} = \boxed{-\frac{5}{4}}$

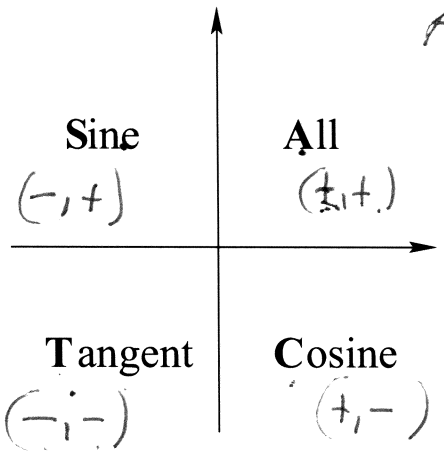
$\cos\theta = \frac{x}{r} = \frac{3}{5}$ $\sec\theta = \frac{r}{x} = \boxed{\frac{5}{3}}$

$\tan\theta = \frac{y}{x} = \boxed{\frac{-4}{3}}$ $\cot\theta = \frac{x}{y} = \boxed{-\frac{3}{4}}$

Checkpoint: Lecture 27, problem 4

Signs of the Trigonometric Functions

All Students Take Calculus

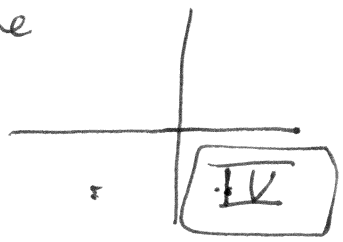


ex. Find the quadrant in which θ lies if $\tan\theta < 0$ and $\sin\theta < 0$

$\tan\theta = \frac{y}{x} = \frac{-}{+}$

y is negative

x is positive

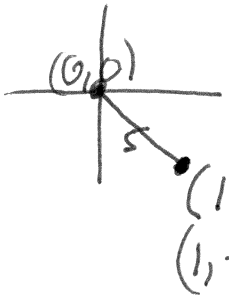


(+, −)

ex. Given $\sec\theta = 5$ and $\sin\theta < 0$, find $\csc\theta$.

(1) $\sec\theta = 5 = \frac{5}{1} = \frac{r}{x}$ so $x = 1$, $r = 5$, also y is negative, so Q IV.

(2) $d = r = 5 = \sqrt{(1-0)^2 + (y-0)^2}$



$5 = \sqrt{1+y^2}$

$25 = 1 + y^2$

$y^2 = 24$

$y = \pm\sqrt{24}$

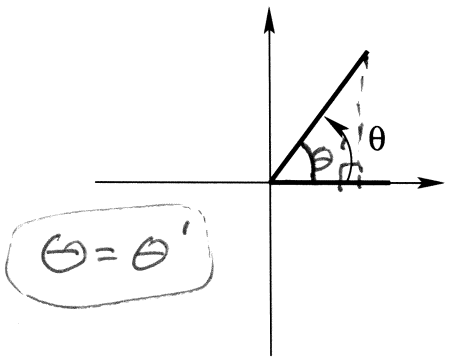
$y = -2\sqrt{6}$

$\csc\theta = \frac{r}{y} = \frac{5}{-2\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}}$

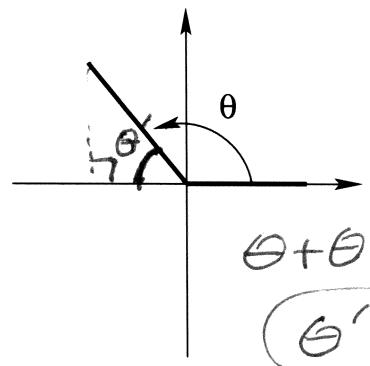
$= \boxed{-\frac{5\sqrt{6}}{12}}$

Reference Angles

<u>Def.</u> Let θ be an angle in standard position. The **reference angle** is the acute angle θ' formed by the terminal side of θ and the x-axis.

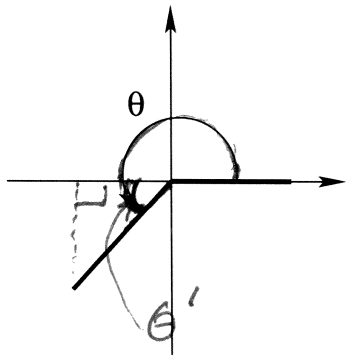


$\theta = \theta'$



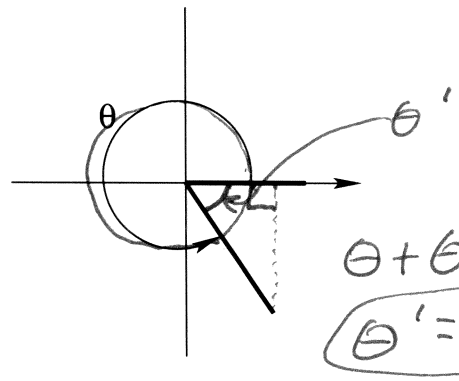
$\theta + \theta' = \pi$

$\theta' = \pi - \theta$



$\theta = \pi + \theta'$

$\theta' = \theta - \pi$



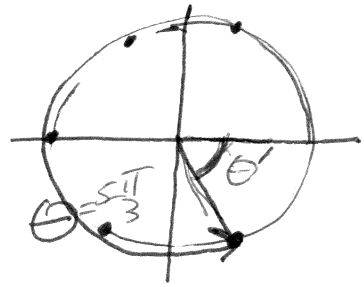
$\theta + \theta' = 2\pi$

$\theta' = 2\pi - \theta$

ex. Find the reference angle for

1) $\theta = \dfrac{5\pi}{3}$

$5\left(\frac{\pi}{3}\right)$



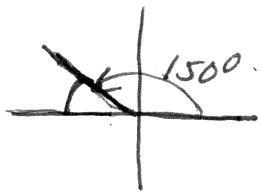
$\theta + \theta' = 2\pi$

$\theta' = 2\pi - \theta$

$\theta' = 2\pi - \frac{5\pi}{3} = \boxed{\frac{\pi}{3}}$

2) $\theta = 870°$ Find the smallest positive coterminal angle that is less than 360°.

$\begin{array}{r} -720 \\ \hline 150° \end{array}$ 720



$\boxed{30°}$

Checkpoint: Lecture 27, problem 5

Evaluating Trigonometric Functions for Any Angle

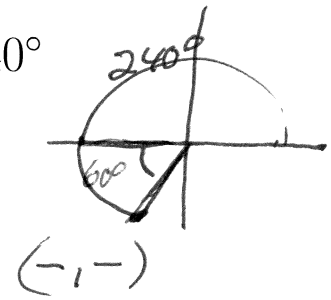
1. Find the reference angle θ' associated with the angle θ.

2. Determine the sign of the trigonometric function of θ by noting the quadrant in which θ lies.

3. The value of the trigonometric function of θ is the same, except possibly for sign, as the value of the trigonometric function of θ'.

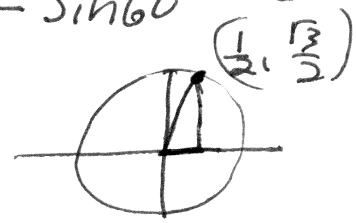
ex. Use the reference angle to find

1) $\sin 240°$

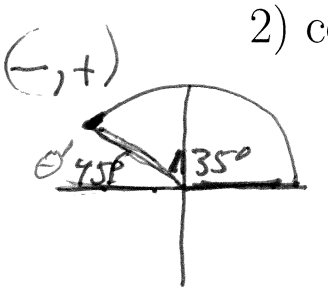


$$\begin{array}{r} 240 \\ 180 \\ \hline 60 \end{array}$$

$\sin 240° = -\sin 60° = \boxed{-\frac{\sqrt{3}}{2}}$



2) $\cot 495° = -$



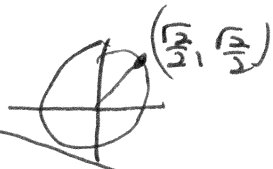
$$\begin{array}{r} -360 \\ \hline 135 \end{array} \qquad \begin{array}{r} 180° \\ -135 \\ \hline 45° \end{array}$$

$\cot 495° = +\cot 135° =$

$= -\cot 45° = -\frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}}$

$= \cancel{-\frac{\sqrt{2}}{2}}$

$= \boxed{-1}$

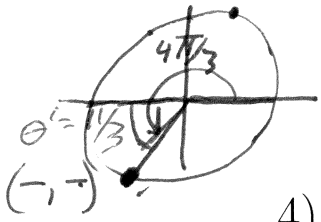


$\frac{16}{3}\pi = (5\frac{1}{3})\pi$

$= (4 + 1\frac{1}{3})\pi$

3) $\sin\left(\frac{16\pi}{3}\right) = \sin\frac{4\pi}{3} = -\sin\frac{\pi}{3} = \boxed{-\frac{\sqrt{3}}{2}}$

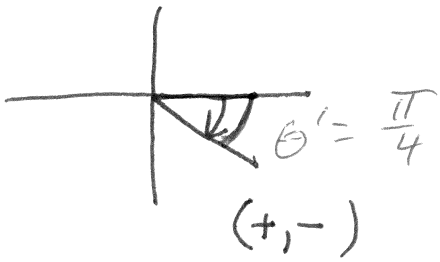
$\frac{16\pi}{3} - 4\pi = \frac{16\pi}{3} - \frac{12\pi}{3} = \frac{4\pi}{3}$



$\frac{1}{\cos(-\frac{\pi}{4})}$

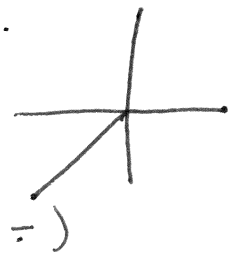
4) $\sec\left(-\frac{\pi}{4}\right) = \sec\frac{\pi}{4} = \frac{1}{\cos\frac{\pi}{4}} = \frac{1}{\frac{\sqrt{2}}{2}} = \frac{2}{\sqrt{2}}\left(\frac{\sqrt{2}}{\sqrt{2}}\right)$

$= \frac{2\sqrt{2}}{2} = \boxed{\sqrt{2}}$



ex. If $\tan\theta = \dfrac{2}{3}$ and θ is in Quadrant III, find $\cos\theta$.

Method 1: Use trigonometric identities. .



$$\tan^2\theta + 1 = \sec^2\theta$$

$$\left(\frac{2}{3}\right)^2 + 1 = \sec^2\theta$$

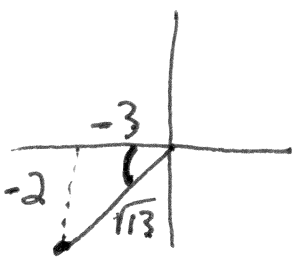
$$\frac{13}{9} = \sec^2\theta$$

$$\sec\theta = \pm\sqrt{\frac{13}{9}}$$

$$\sec\theta = -\frac{\sqrt{13}}{3}$$

$$\cos\theta = -\frac{3}{\sqrt{13}}\,\frac{\sqrt{13}}{\sqrt{13}} = \boxed{-\frac{3\sqrt{13}}{13}}$$

Method 2: Use the reference angle.



$$\tan\theta = \frac{2}{3} = \frac{\text{opp}}{\text{adj}}$$

$$4 + 9 = \text{hyp}^2$$

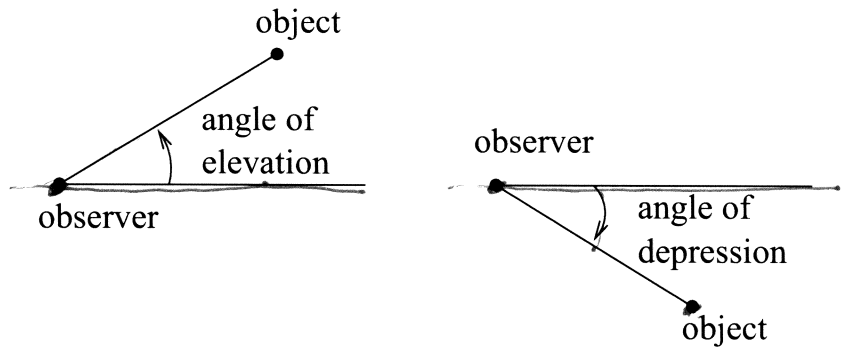
$$13 = \text{hyp}^2$$

$$h = \sqrt{13}$$

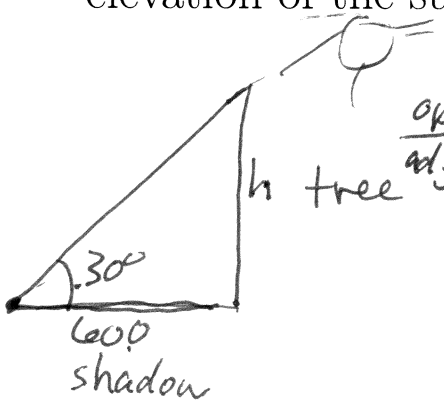
$$\cos\theta = \frac{\text{adj}}{\text{hyp}} = \frac{-3}{\sqrt{13}} = \boxed{-\frac{3\sqrt{13}}{13}}$$

Checkpoint: Lecture 27, problem 6

Applications



ex. A giant redwood tree casts a shadow 600 feet long. Find the height of the tree if the angle of elevation of the sun is 30°.



$$\frac{opp}{adj} = \tan 30^\circ = \frac{h}{600}$$

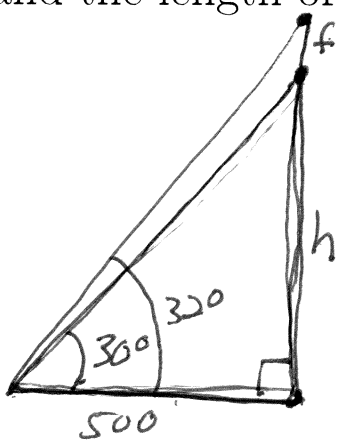
$$h = 600 \tan 30^\circ$$

$$h = 600\left(\frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) = 600 \cdot \frac{1}{\sqrt{3}}\left(\frac{\sqrt{3}}{\sqrt{3}}\right)$$

$$= \frac{600\sqrt{3}}{3} = 200\sqrt{3} \text{ ft}$$

ex. From a point on the ground 500 feet from the base of a building, an observer finds that the angle of elevation to the top of the building is 30° and that the angle of elevation to the top of a flagpole atop the building is 32°. Find the height of the building and the length of the flagpole.

start with smaller Δ



building $\frac{opp}{adj} = \tan 30° = \frac{h}{500}$

$h = 500 \tan 30°$

larger Δ

$\frac{opp}{adj} = \frac{f+h}{500} = \tan 32°$

$f+h = 500 \tan 32°$

$f = 500 \tan 32° - h$

$f = 500 \tan 32° - h$

$f = \left(500 \tan 32° - \frac{500\sqrt{3}}{3}\right) ft$

need calc.

$h = 500 \tan 30°$

$h = 500 \frac{\sqrt{3}}{3} ft$